宏基舜达
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STRUCTURAL STEEL vs REINFORCED CONCRETE

20-year parameter matrix - structural steel vs reinforced concrete

10,000 m^2 (107,640 sq ft) logistics hub



1 DEAD LOAD

RCC 180 kg/m2 (36.9 lb/ft2)

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Steel 32 kg/m2 (6.6 lb/ft2)

Footing concrete 0.22 to 0.09 m3/m2

Excavation 2.5-3.5 m to 1.2-1.8 m

(8.2-11.5 ft to 3.9-5.9 ft)

2 PROGRAMME

Concrete 10-14 months

Steel 4-6 months

At a typical rental yield, five months of earlier handover is a material amount of income that does not have to be financed.

3 SPAN

RCC: columns beyond 24 m (78.7 ft)

Steel clear span 30 / 36 / 48 m

(98.4 / 118.1 / 157.5 ft)

+8-12% usable storage volume, same footprint



Epoxy/polyurethane, 160 um: 15-20 years

Hot-dip galvanising, 85 um: 25+ years

> Concrete: floor joints re-caulked every 3-5 years

CONCRETE WINS THE SLAB. ABOVE THE SLAB, THE NUMBERS FLIP.

Is your tender scored on first cost - or 20-year cost?

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